Chapter 10: Hypothesis Testing

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Hypothesis Testing when σ is Known

We now turn to a procedure known as **hypothesis testing**.

To properly appreciate this, you must remember what the principle issue is when we conduct statistical analysis. As a researcher, we **often do not know** what the value of μ is. This means that we have no choice by to rely on our sample estimate $\bar{x}$.

In the previous chapter, our objective was to **estimate the unknown population mean**. In hypothesis testing, however, we are interested in **evaluating a specific claim about the population mean**.

Hypothesis Testing when σ is Known

Suppose the Bahraini Ministry of Labour wants to investigate the average number of employees working at companies in Bahrain.

A researcher collects a random sample of 25 companies and records the number of employees working at each company.

The sample produces the following information:

$$n = 25$$
$$\bar{x} = 12$$

Hypothesis Testing when σ is Known

This shows that the average number of employees per company in our sample is 12.

However, since this is only a sample, we cannot immediately conclude that the true population mean is also 12.

Suppose that the Minister of Education instead believes that:

$$\mu = 10$$

Hypothesis Testing when σ is Known

Recall that different random samples can produce different sample means, even when they are drawn from the same population.

Now what we are interested in asking is if the true population mean is $\mu = 10$, how unusual would it be to observe a sample mean of $\bar{x} = 12$?

Hypothesis testing provides a statistical framework for determining whether an observed sample result is sufficiently unusual to reject a particular claim about the population.

The Null and Alternative Hypotheses

The first step in conducting a hypothesis test is to specify two competing hypotheses, which we call the null hypothesis H_0 and the alternative hypothesis H_1.

The null hypothesis represents the claim that we wish to evaluate. In our example:

$$H_0 : \mu = 10$$

When conducting a hypothesis test, we temporarily assume that the null hypothesis is true.

The Null and Alternative Hypotheses

The alternative hypothesis represents the competing claim. For example, if we are interested in whether the population mean is greater than 10, we would specify:

$$H_0 : \mu = 10$$
$$H_1 : \mu > 10$$

The Null and Alternative Hypotheses

The choice of alternative hypothesis depends on the research question being investigated and your selection will change the way to we conduct the test.

We will discuss this shortly, but for now let us focus on the null hypothesis that $\mu = 10$ and examine how unusual our observed sample mean would be under this assumption.

When σ is Known

Suppose we are told that the population standard deviation of the number of employees per company is:

$$\sigma = 5$$

We therefore have the following information:

$$\begin{aligned} n &= 25 \\ \bar{x} &= 12 \\ \sigma &= 5 \\ \mu_0 &= 10 \end{aligned}$$

where μ_0 represents the population mean specified under the null hypothesis.

When σ is Known

Since we have a random sample of 25 companies, suppose that the population distribution is sufficiently well behaved for the Central Limit Theorem to provide a reasonable normal approximation.

Under the null hypothesis:

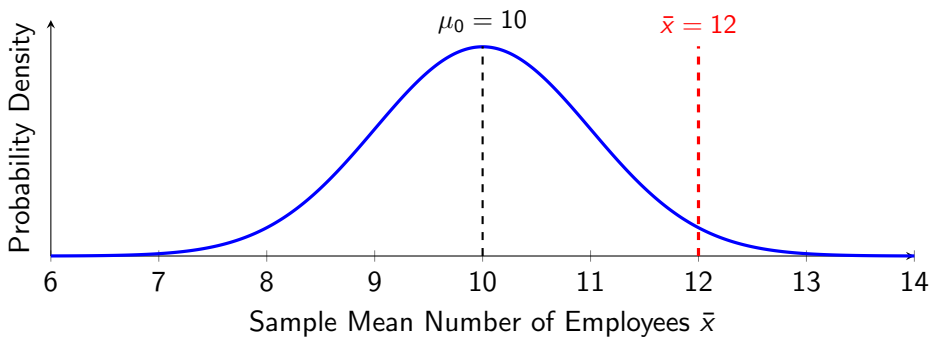
$$\bar{X} \overset{\text{approx}}{\sim} N\left(\mu_0, \frac{\sigma^2}{n}\right)$$

$$\bar{X} \overset{\text{approx}}{\sim} N(10, 1)$$

When σ is Known

We can illustrate this distribution as follows:

Figure 1: Sampling Distribution of the Sample Mean Under the Null Hypothesis



When σ is Known

Notice that the observed sample mean is larger than the hypothesised population mean.

However, the important question is not simply whether $\bar{x}$ differs from μ_0. Instead, we want to determine **how far the observed sample mean is from the hypothesised population mean**.

The Test Statistic

To measure how unusual our sample mean is, we need to start by standardising the sampling distribution.

Recall that the standardisation formula for an individual observation is:

$$Z = \frac{X - \mu}{\sigma}$$

Since the standard deviation of the sampling distribution is $\frac{\sigma}{\sqrt{n}}$, we instead use:

The Test Statistic

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

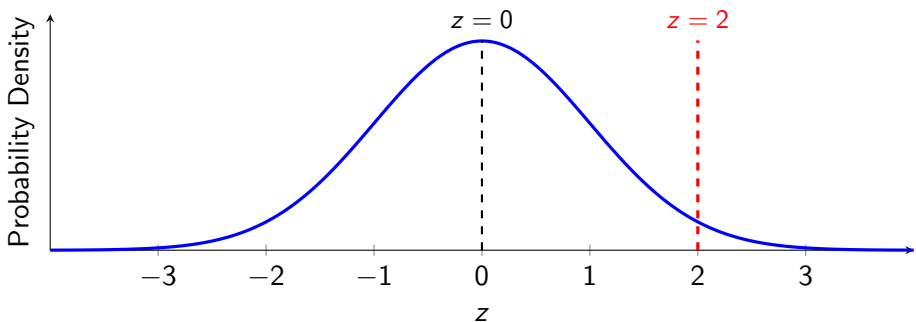
Substituting our values:

$$\begin{aligned} z &= \frac{12 - 10}{5/\sqrt{25}} \\ &= \frac{2}{1} \\ &= 2 \end{aligned}$$

The Test Statistic

This tells us that the observed sample mean $\bar{x} = 12$ is **two standard errors away from** the population mean specified under the null hypothesis.

Figure 2: Standardised Sampling Distribution Under the Null Hypothesis



The Test Statistic

Now we are interested in the following question: is 2 standard deviations sufficiently far enough from $\mu = 10$ that we can reject the null hypothesis H_0?

The **decision rule** is to simply ensure that your z-value for $\bar{x}$ must be greater than α:

$$z > \alpha$$

In this case:

$$2 > \alpha$$

The Test Statistic

In order to determine what the value of α is, we will need to know what the **alternative hypothesis** H_1 is and what the **significance level** of our test will be. It will be one of the values that come from the below table:

The Test Statistic

Table 1: Critical Values of the t-Distribution for Hypothesis Testing

	Level of Significance for One-Tailed Test						
	20%	**15%**	**10%**	**5%**	**2.5%**	**1%**	**0.5%**
	Level of Significance for Two-Tailed Test						
df	**40%**	**30%**	**20%**	**20%**	**5%**	**2%**	**1%**
1	1.376	1.963	3.078	6.314	12.706	31.821	63.657
2	1.061	1.386	1.886	2.920	4.303	6.965	9.925
3	0.978	1.250	1.638	2.353	3.182	4.541	5.841
4	0.941	1.190	1.533	2.132	2.776	3.747	4.604
5	0.920	1.156	1.476	2.015	2.571	3.365	4.032
6	0.906	1.134	1.440	1.943	2.447	3.143	3.707

The Test Statistic

Table 2: Critical Values of the t-Distribution for Hypothesis Testing

	Level of Significance for One-Tailed Test						
	20%	**15%**	**10%**	**5%**	**2.5%**	**1%**	**0.5%**
	Level of Significance for Two-Tailed Test						
df	**40%**	**30%**	**20%**	**20%**	**5%**	**2%**	**1%**
7	0.896	1.119	1.415	1.895	2.365	2.998	3.499
8	0.889	1.108	1.397	1.860	2.306	2.896	3.355
9	0.883	1.100	1.383	1.833	2.262	2.821	3.250
10	0.879	1.093	1.372	1.812	2.228	2.764	3.169
∞	0.842	1.036	1.282	1.645	1.960	2.326	2.576

One-Tailed Test

Suppose we test the below hypothesis:

$$H_0 : \mu = 10$$
$$H_1 : \mu > 10$$

Essentially, we are saying that if we reject the null hypothesis H_0, then there is sufficient evidence supporting the alternative must, which is that μ is some number that is greater than 10.

One-Tailed Test

This is known as a **one-tailed test**. This is because if $\mu > 10$, then we are only testing half of the distribution (i.e, only the right tail):

Recall that our observed Z-statistic is:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

$$= \frac{12 - 10}{5/\sqrt{25}}$$

$$= 2$$

One-Tailed Test

In order for us to reject the null $H_0 : \mu = 10$, we need the z-value of $\bar{x}$ to be greater than the critical value:

$$z > z_\alpha$$

In a one tailed test when the value of σ is known, we only need to observe the row for ∞ to get the value of α:

One-Tailed Test

Table 3: Critical Values of the t-Distribution for Hypothesis Testing

	Level of Significance for One-Tailed Test						
	20%	**15%**	**10%**	**5%**	**2.5%**	**1%**	**0.5%**
∞	0.842	1.036	1.282	1.645	1.960	2.326	2.576

One-Tailed Test

The percentages on each column represents the **significance level** α of the test. The smaller the percentage, **the more plausible** your results become.

This is because the significance level represents the chance that you are wrong when rejecting the null hypothesis.

For example, a significance level of 10% means that, if the null hypothesis is true, our testing procedure has a 10% probability of incorrectly rejecting it.

One-Tailed Test

In order to reject the null hypothesis, we need to check:

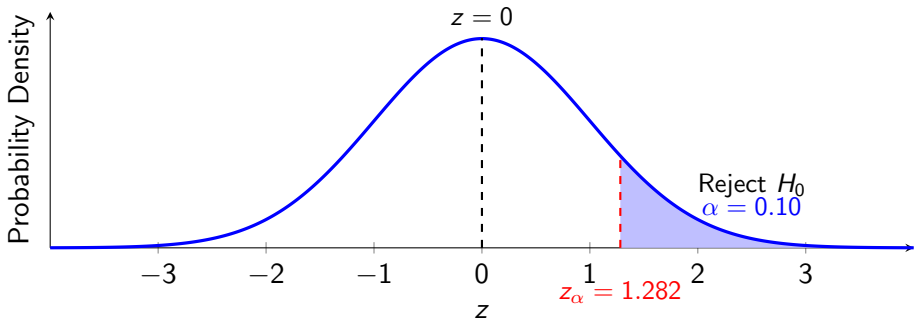
$$z > z_{10\%}$$
$$2 > 1.282$$

This shows that $\bar{x} = 12$ is indeed statistically different from $\mu = 10$. We reject the null hypothesis and accept the alternative H_1 that $\mu > 10$.

Graphically, this is shown as follows:

One-Tailed Test

Figure 3: Right-Tailed Hypothesis Test



One-Tailed Test

Intuitively, this means that $\bar{x} = 12$ is indeed an unusually large number compared to $\mu = 10$ based on how many standard deviations away it is from the centre.

We would therefore conclude that μ is some number that is greater than 10.

One-Tailed Test

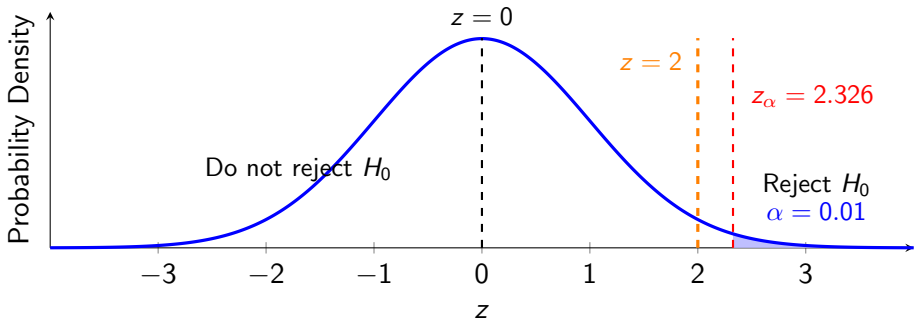
Now let us instead test our hypothesis with a 1% significance level. This means that to reject the null:

$$z > z_{1\%}$$
$$2 < 2.326$$

In this case, there is **insufficient evidence** to reject the null hypothesis. It is therefore plausible that $\mu = 10$:

One-Tailed Test

Figure 4: Right-Tailed Hypothesis Test ($\alpha = 0.01$)



Two-Tailed Test

Suppose now we test the below hypothesis:

$$H_0 : \mu = 10$$
$$H_1 : \mu \neq 10$$

Notice this time the alternative hypothesis does not make any assumptions about whether the value of $\bar{x}$ is over or under 10. It is simply saying that it is not equal to 10.

Two-Tailed Test

This is known as a two-tailed test. This is because we are essentially testing the hypothesis **under both tails of the distribution**.

The decision rule for rejection is still the same. We simply have to select the z_α which corresponds for a two-tailed test:

Two-Tailed Test

Table 4: Critical Values of the t-Distribution for Hypothesis Testing

	Level of Significance for One-Tailed Test						
	20%	**15%**	**10%**	**5%**	**2.5%**	**1%**	**0.5%**
	Level of Significance for Two-Tailed Test						
∞	0.842	1.036	1.282	1.645	1.960	2.326	2.576

Two-Tailed Test

Suppose we wanted to test the hypothesis under the 5% significance level. We would simply select $z_{5\%} = 1.960$:

$$z > z_{5\%}$$
$$2 > 1.96$$

We would therefore reject the null hypothesis that $H_0 : \mu = 10$.

Hypothesis Testing when σ is Unknown

Now let us consider when the population variance σ was unknown to us.

Just as before, we would have to instead use the sample estimator s instead.

However the consequence of doing this is that we are no longer able to say that $\bar{x}$ has a normal distribution. We instead say that it follows the t-distribution:

$$\bar{x} \sim t_{n-1}$$

Hypothesis Testing when σ is Unknown

Remember that the t-distribution still has the same symmetric shape that a normal distribution has. For a sufficiently large sample $n = \infty$, our t-distribution approximates a normal distribution. The procedure for hypothesis testing is slightly different in this circumstance.

Recall that when σ is known, we use the Z-test statistic:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Hypothesis Testing when σ is Unknown

When σ is unknown, we replace it with the sample standard deviation s:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

This is known as the **t-test statistic**.

Hypothesis Testing when σ is Unknown

Suppose a researcher wants to investigate whether the average height of university students in Bahrain exceeds 170 cm.

The researcher collects a random sample of 10 students and obtains the following information:

$$n = 10$$
$$\bar{x} = 174 \text{ cm}$$
$$s = 6 \text{ cm}$$

Hypothesis Testing when σ is Unknown

The researcher's hypothesis becomes:

$$H_0 : \mu \leq 170$$
$$H_1 : \mu > 170$$

To calculate the t-statistic:

Hypothesis Testing when σ is Unknown

The formula is:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$t = \frac{174 - 170}{6/\sqrt{10}}$$

$$= \frac{4}{1.897}$$

$$= 2.108$$

Hypothesis Testing when σ is Unknown

Let us conduct our test at the 5% significance level. In this case, we would need to look at the table and select the row where $df = 10 - 1 = 9$:

Hypothesis Testing when σ is Unknown

Table 5: Critical Values of the t-Distribution for Hypothesis Testing

	Level of Significance for One-Tailed Test						
	20%	**15%**	**10%**	**5%**	**2.5%**	**1%**	**0.5%**
	Level of Significance for Two-Tailed Test						
df	**40%**	**30%**	**20%**	**20%**	**5%**	**2%**	**1%**
8	0.889	1.108	1.397	1.860	2.306	2.896	3.355
9	0.883	1.100	1.383	1.833	2.262	2.821	3.250
10	0.879	1.093	1.372	1.812	2.228	2.764	3.169
∞	0.842	1.036	1.282	1.645	1.960	2.326	2.576

Hypothesis Testing when σ is Unknown

The corresponding critical value is:

$$t_{0.05,9} = 1.833$$

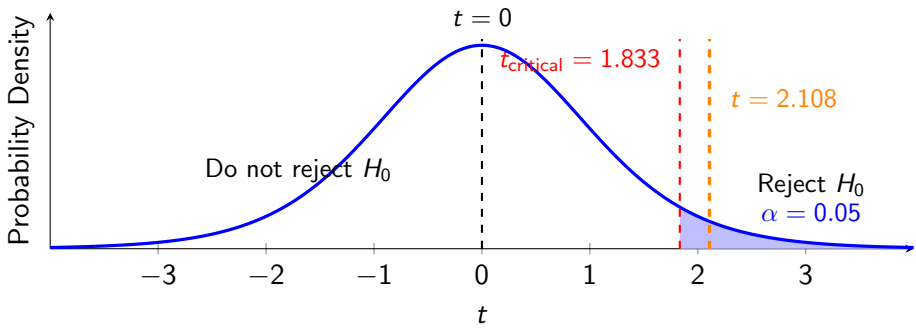
For a right-tailed t-test, the decision rule is:

$$t > t_{\alpha,df}$$
$$2.108 > 1.833$$

Hypothesis Testing when σ is Unknown

Since our calculated t-statistic exceeds the critical value, we **reject the null hypothesis**.

Figure 5: Right-Tailed t-Test ($df = 9$, $\alpha = 0.05$)



Hypothesis Testing when σ is Unknown

Therefore, at the 5% significance level, there is sufficient statistical evidence to conclude that the population mean height of male university students exceeds 170 cm.

Hypothesis Testing with Proportional Variables

What we have discussed so far relates to testing hypotheses when the random variable X is continuous. When analysing discrete variables, we would need to use the appropriate estimators.

Recall from the previous chapter that a proportion and its variance is defined as:

$$\pi = \frac{\sum x_i}{N} \qquad p = \frac{x}{n}$$

Hypothesis Testing with Proportional Variables

The variance and standard deviation is given by:

$$\sigma^2 = \frac{\pi(1-\pi)}{n} \qquad \sigma = \sqrt{\frac{\pi(1-\pi)}{n}}$$

In this module, we only examine the case when σ is known.

In order to conduct a hypothesis test for discrete variables, the value of z for our decision rule will have the below formula:

$$Z = \frac{|p-\pi|}{\sqrt{\frac{\pi(1-\pi)}{n}}}$$

Hypothesis Testing with Proportional Variables

Activity:

A large retail company in Bahrain aims to maintain a customer satisfaction rate of at least 80%. The company's management is concerned that customer satisfaction may have fallen below this target.

To investigate this, the company conducts a random survey of 2,000 customers. The survey reveals that 77.5% of customers are satisfied with the company's services.

Using a **5% significance level**, conduct a hypothesis test to determine whether there is sufficient statistical evidence that the population proportion of satisfied customers has fallen below 80%.

Solution

We are given the following information:

$$n = 2000$$
$$p = 0.775$$
$$\pi_0 = 0.80$$

Hypothesis Testing with Proportional Variables

We want to determine whether the population proportion of satisfied customers is less than 80%.

$$H_0 : \pi \geq 0.80$$
$$H_1 : \pi < 0.80$$

This is a **left-tailed hypothesis test** because the alternative hypothesis states that the population proportion is less than 0.80.

Hypothesis Testing with Proportional Variables

We are conducting our hypothesis test at the 5% significance level:

$$\alpha = 0.05$$

Since this is a left-tailed Z-test, the corresponding critical value is:

$$z_\alpha = -1.645$$

We need to calculate the z-value:

Hypothesis Testing with Proportional Variables

$$
\begin{aligned}
z &= \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \\
&= \frac{0.775 - 0.80}{\sqrt{\frac{0.80(1-0.80)}{2000}}} \\
&= \frac{-0.025}{\sqrt{\frac{0.80(0.20)}{2000}}} \\
&= \frac{-0.025}{\sqrt{0.00008}} \\
&= \frac{-0.025}{0.008944} \\
&= -2.80
\end{aligned}
$$

Hypothesis Testing with Proportional Variables

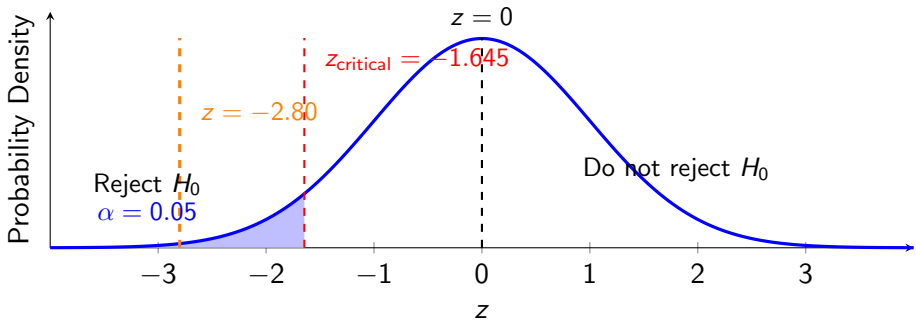
For a left-tailed hypothesis test, we reject the null hypothesis if the calculated Z-statistic is smaller than the critical value:

Where we can see:

$$-2.80 < -1.645$$

Hypothesis Testing with Proportional Variables

Figure 6: Left-Tailed Z-Test for a Population Proportion ($\alpha = 0.05$)



Our observed Z-statistic of -2.80 falls inside the rejection region. Therefore, we reject H_0.

Hypothesis Testing for Paired Samples

Consider an example where a company in Bahrain introduces a new training programme to improve employee productivity.

The company wants to investigate whether its employees become more productive after completing the programme.

One approach would be to measure the productivity of the **same employees** before and after receiving the training.

Hypothesis Testing for Paired Samples

This is an example of **paired samples**. That is, when two observations are collected from the same individual.

The objective of a paired-sample hypothesis test is to determine whether there is a statistically significant difference between the two measurements.

Hypothesis Testing for Paired Samples

Let us analyse the results of this training programme:

Table 6: Employee Productivity Before and After Training

Employee i	**Before Training**	**After Training**	**Difference d_i**
1	20	25	5
2	25	28	3
3	18	24	6
4	22	24	2
5	30	34	4

Hypothesis Testing for Paired Samples

Notice that each employee has two observations. For example, Employee 1 completed

20 tasks before training and 25 tasks after training.

Therefore, their productivity increased by 5 tasks. Similarly, Employee 2 increased their productivity by 3 tasks.

The central idea behind paired-sample hypothesis testing is that we transform our two measurements into a single variable representing their difference.

Hypothesis Testing for Paired Samples

Let:

$$X_{i,1} = \text{Productivity before training}$$
$$X_{i,2} = \text{Productivity after training}$$

We define the difference for employee i as:

$$D_i = X_{i,2} - X_{i,1}$$

Hypothesis Testing for Paired Samples

For example, the difference for Employee 1 is:

$$\begin{aligned} d_1 &= 25 - 20 \\ &= 5 \end{aligned}$$

The difference for Employee 2 is:

$$\begin{aligned} d_2 &= 28 - 25 \\ &= 3 \end{aligned}$$

Hypothesis Testing for Paired Samples

Therefore, instead of working with two separate columns of productivity measurements, we can work with a single variable containing the differences:

$$d = \{5, 3, 6, 2, 4\}$$

A **paired-sample hypothesis test** is essentially a **one-sample hypothesis test** applied to the differences between paired observations.

Once we calculate the differences, we can calculate their sample mean.

Hypothesis Testing for Paired Samples

The formula is:

$$\bar{d} = \frac{\sum_{i=1}^{n} d_i}{n}$$

where:

- $\bar{d}$ is the sample mean difference.
- d_i is the difference for observation i.
- n is the number of paired observations.

Hypothesis Testing for Paired Samples

Using our example:

$$\bar{d} = \frac{5+3+6+2+4}{5}$$

$$= \frac{20}{5}$$

$$= 4$$

Therefore, employee productivity increased by an average of 4 tasks following the training programme.

Hypothesis Testing for Paired Samples

However, this does not necessarily mean that the improvement is statistically significant. As usual, the hypothesis test would measure how many standard deviations away we are from the the null.

We specify the following hypotheses:

$$H_0 : \mu_D \leq 0$$
$$H_1 : \mu_D > 0$$

Hypothesis Testing for Paired Samples

The null hypothesis states that there is no positive average change in employee productivity following the training programme. The alternative hypothesis states that the population mean difference is positive.

Recall that when the population standard deviation is unknown, we use the t-test statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Hypothesis Testing for Paired Samples

For paired samples, we use almost the same formula:

$$t = \frac{\bar{d} - \mu_{D,0}}{s_D/\sqrt{n}}$$

where:

- $\bar{d}$ is the sample mean difference.
- $\mu_{D,0}$ is the hypothesised population mean difference, usually zero.
- s_D is the sample standard deviation of the differences.
- n is the number of paired observations.

Hypothesis Testing for Paired Samples

We previously calculated:

$$\bar{d} = 4$$

Using a **5% significance level**, conduct a paired-sample hypothesis test to determine whether employee productivity has increased following the training programme.

Recall that the sample variance is:

$$s_D^2 = \frac{\sum_{i=1}^{n}(d_i - \bar{d})^2}{n-1}$$

Hypothesis Testing for Paired Samples

Substituting our values:

$$
\begin{aligned}
s_D^2 &= \frac{(5-4)^2 + (3-4)^2 + (6-4)^2 + (2-4)^2 + (4-4)^2}{5-1} \\
&= \frac{1+1+4+4+0}{4} \\
&= \frac{10}{4} \\
&= 2.5
\end{aligned}
$$

Hypothesis Testing for Paired Samples

Therefore, the sample standard deviation of the differences is:

$$\begin{aligned} s_D &= \sqrt{2.5} \\ &= 1.581 \end{aligned}$$

The paired-sample t-statistic is:

$$t = \frac{\bar{d} - \mu_{D,0}}{s_D/\sqrt{n}}$$

Hypothesis Testing for Paired Samples

Substituting our values:

$$\begin{aligned} t &= \frac{4 - 0}{1.581/\sqrt{5}} \\ &= \frac{4}{0.7071} \\ &\approx 5.657 \end{aligned}$$

Therefore, the calculated test statistic is:

$$t = 5.657$$

Hypothesis Testing for Paired Samples

We are conducting a right-tailed hypothesis test at the 5% significance level:

From the t-distribution table, the critical value for a one-tailed test with $df = 4$ and $\alpha = 0.05$ is:

$$t_{0.05,4} = 2.132$$

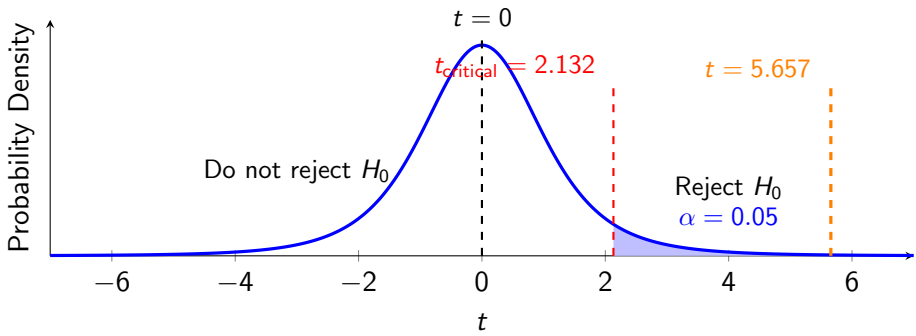
Hypothesis Testing for Paired Samples

For a right-tailed hypothesis test, we reject the null hypothesis if:

$$t > t_{\alpha,df}$$
$$5.657 > 2.132$$

Hypothesis Testing for Paired Samples

Figure 7: Paired-Sample Right-Tailed t-Test ($df = 4$, $\alpha = 0.05$)



Since our calculated t-statistic exceeds the critical value, we **reject the null hypothesis**.